

APPLIED MATH 3

WEEK 1

INTRODUCTION

Basic input and movement strategies – 2D

INPUT AND PERSPECTIVE

- Deciding on your input and camera perspective should be one of your first decisions made when designing a new game prototype.
- This is because the input layout will partially be determined by how the camera follows the player avatar
- If the Camera rotates with the player movement, then the input will have to move the player RELATIVE to the players orientation. (eg 3rd Person Games)
- If the camera is fixed to the world, or is controlled separately, then the input will be relative to the world space

CAMERA PERSPECTIVE CONSIDERATIONS

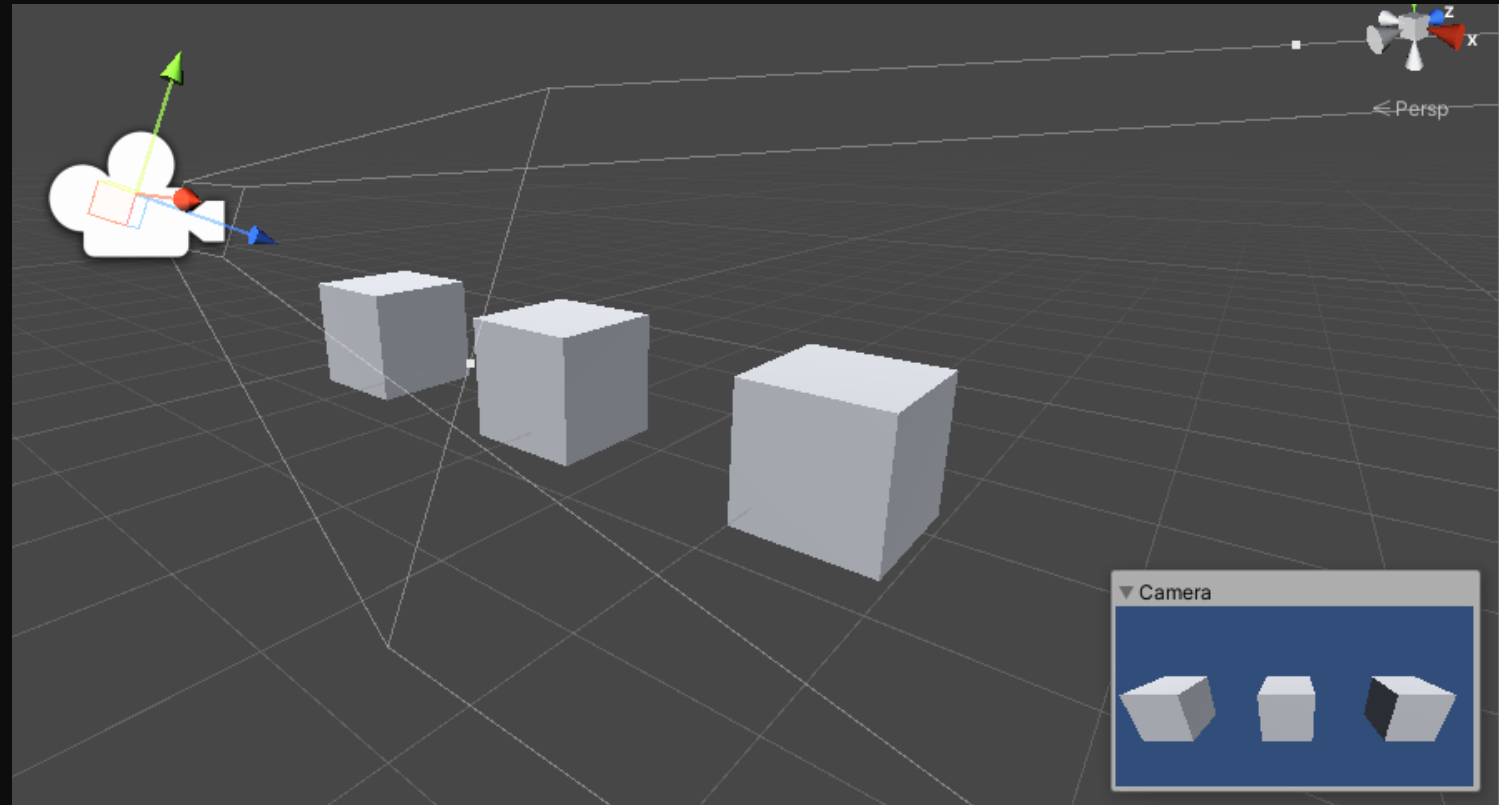
- There are also other considerations regarding art, physics and other factors that will be affected by the camera movement.
 - Does the camera rotate to follow the player orientation, OR is the camera fixed in the world?
 - Is the camera position locked or dynamic?
 - Can the player move the camera independently?

CAMERA PERSPECTIVE CONSIDERATIONS

- While you do have options to change this (somewhat) later, it is an important decision that will have many wide ranging effects on gameplay later on, so it is important to consider it early.
- This is because the input is usually relative to EITHER the player avatar, or the screen space.

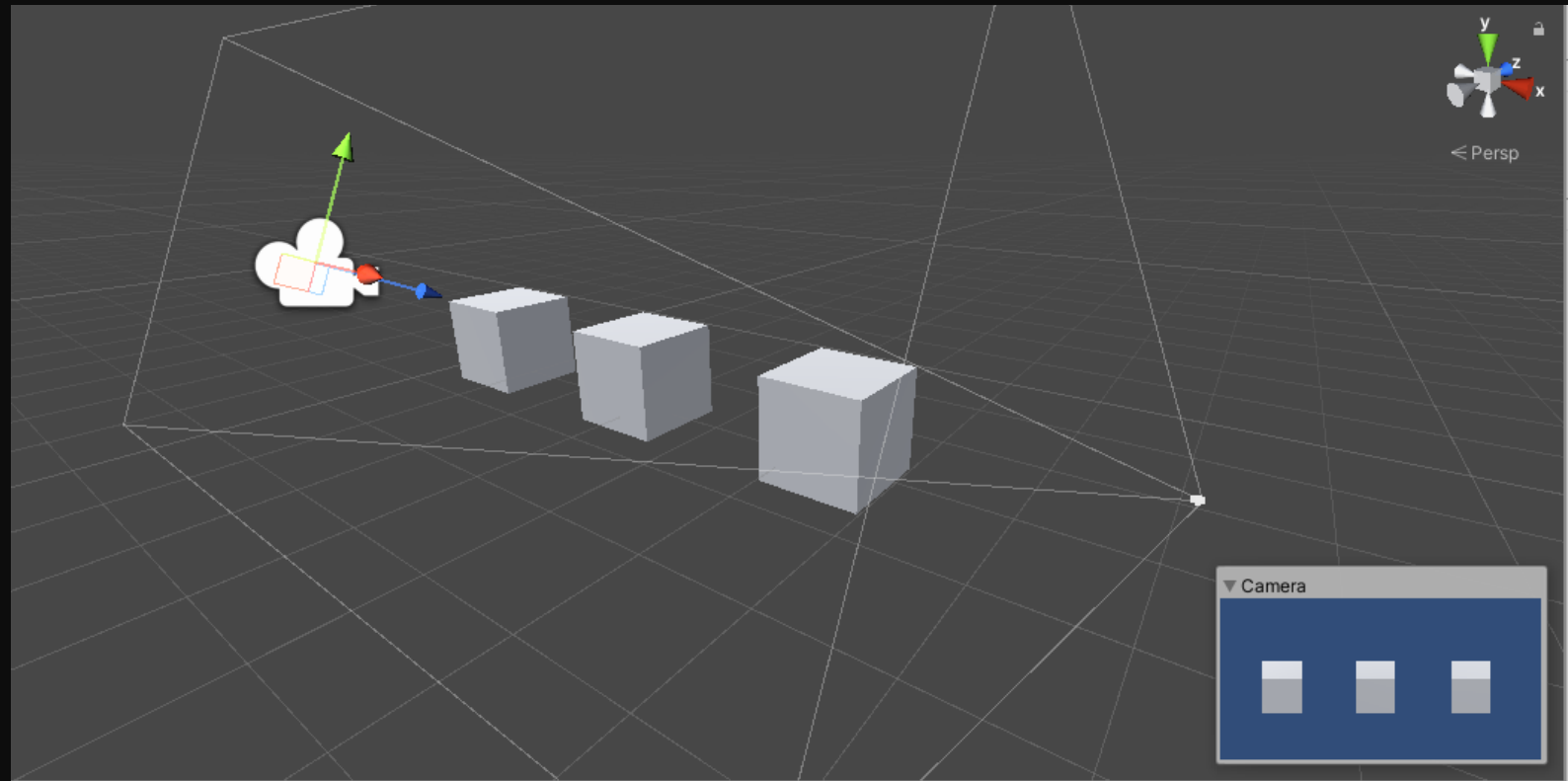
CAMERA PERSPECTIVE TYPES

Perspective Camera: The camera perspective originates from a single point. Objects viewed will be distorted by the perspective. The amount of distortion is different depending on the field of view (width of viewing angle) of the camera. Note how each of the cubes is distorted relative to its position to the camera.



CAMERA PERSPECTIVE TYPES

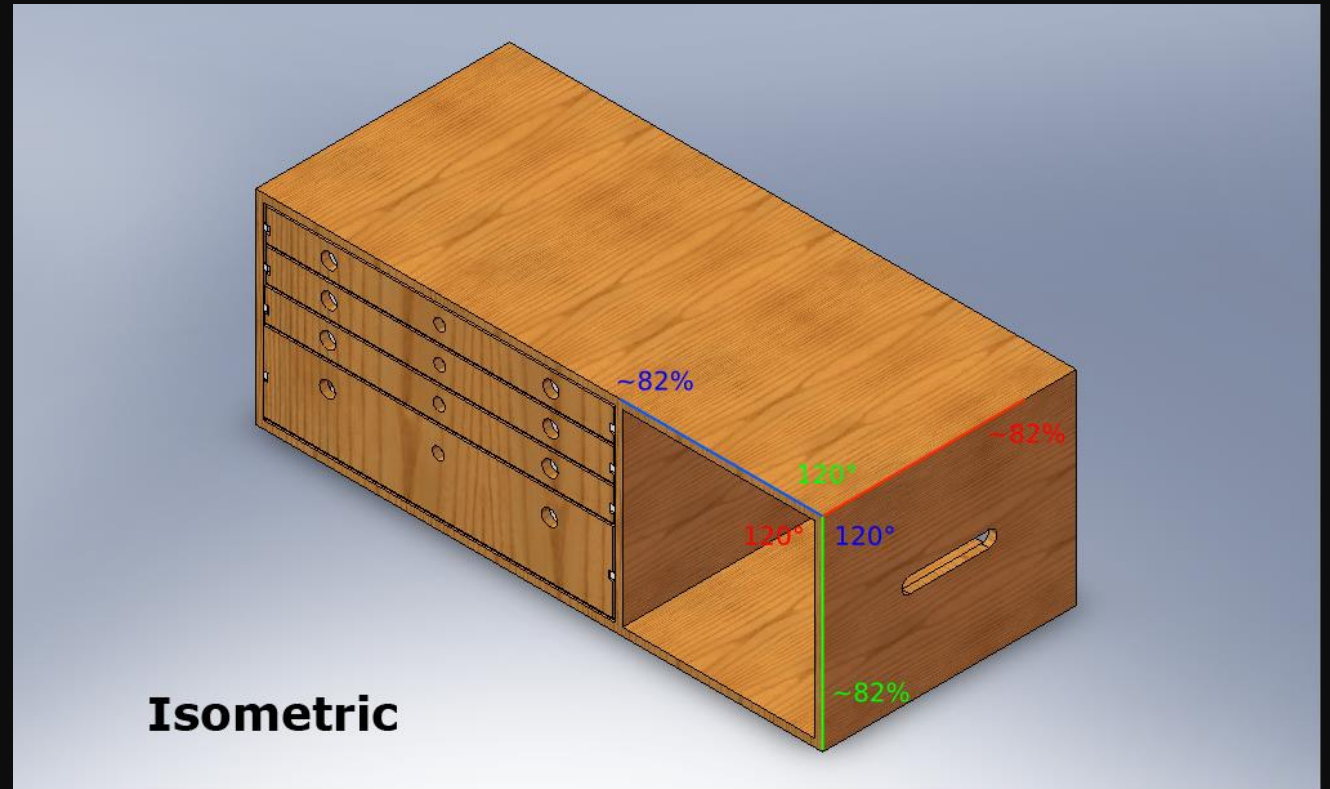
Orthographic Camera: An orthographic camera has no perspective distortion. Note how all three cubes appear exactly the same.



CAMERA PERSPECTIVE TYPES

Isometric projection: Strictly speaking, this has a direction of viewing so that all three axes of space are foreshortened equally, with a common 120° angle between them. This term is often loosely used for anything that is seen from a $\frac{3}{4}$ above perspective.

Image Credit: https://en.wikipedia.org/wiki/3D_projection



CAMERA PERSPECTIVE: FIXED

Fixed perspective:
Game: Full Throttle

In this example, the camera is set on a scene by scene basis and never moves. Player control is simply clicking on objects or people to interact with them.

This is a good perspective for a narrative game with 2D art. It allows for greater control over the emotional tone of each scene.





CAMERA PERSPECTIVE: PLATFORMER (SIDE)

Platformers (typically) use a fixed, side perspective camera.

This is a relatively simple camera to both make and use, making it highly accessible.

There are many variations and tricks but it allows you to use 2D geometry for the level, which also simplifies the NPC behaviour.



+5

Bonus

152

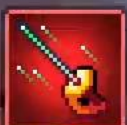
01:46



LB



H



Y



LT



RT



3

3

3

188 / 1078

Seed: 052911

8f91c77b - #052911.V26

2m 43s



CAMERA PERSPECTIVE CONSIDERATIONS

Locked Rotation:

Game: Hades (Supergiant Games)

In this mode, the camera smoothly pans to follow the player, however it never rotates. This allows the use of 2D art assets (or a mix of 2D and 3D) since the perspective is always the same.

This is also an example of an Isometric perspective with an Orthographic camera.



CAMERA PERSPECTIVE CONSIDERATIONS

Locked Rotation:

Game: Hades (Supergiant Games)

In terms of controls, Left on the gamepad will always move the character to the Left in the game. That is, the controls are locked to the camera axis.

This is a very intuitive movement scheme that has been around since early console days and is very approachable for most players.



GIVE TO A FRIEND

Y AXIS

X AXIS

1
10
10
1

CAMERA PERSPECTIVE CONSIDERATIONS

Question: Should movement be applied equally to the X and Y axis of the player?



CAMERA PERSPECTIVE CONSIDERATIONS

Semi Locked Rotation:

Game: Bad North

Studio:

In this mode, the player is able to rotate the camera around a central point, and zoom in and out, but the angle never changes.

This is an example of a non-Isometric $\frac{3}{4}$ perspective. It also appears to be using an Orthographic camera as there is no distortion as the player zooms in or out.

Because it can rotate, 3D assets are required.



CAMERA PERSPECTIVE CONSIDERATIONS

Perspective Camera: The camera perspective originates from a single point. Objects viewed will be distorted by the perspective. The amount of distortion is different depending on the width of the angle of the camera.

Orthographic Camera: An orthographic camera has no perspective distortion.

Isometric projection: Strictly speaking, this has a direction of viewing so that all three axes of space are foreshortened equally, with a common 120° angle between them. This term is often loosely used for anything that is seen from a $\frac{3}{4}$ above perspective.



CAMERA PERSPECTIVE CONSIDERATIONS

Semi Locked Rotation:

Game: Bad North

Studio:

In terms of control, either a mouse or gamepad can be used. In this case, a gamepad sets target positions for your units by moving relative to the camera. This is helped by the cursor snapping to a grid / unit to help select.

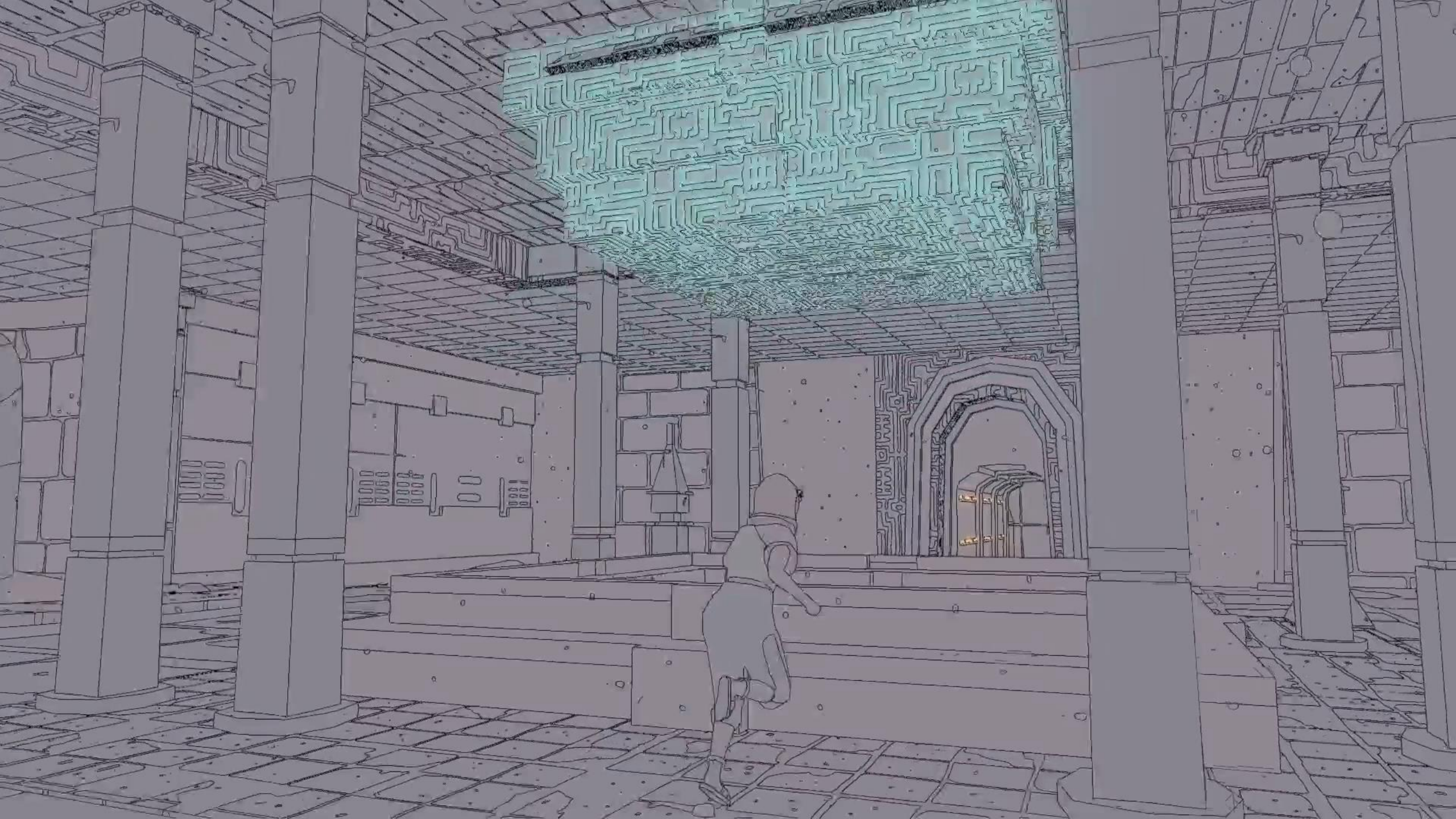
Even the small amount of camera movement requires a substantially more complex control scheme however (although having the grid is beneficial for pathfinding and AI).





Eldred

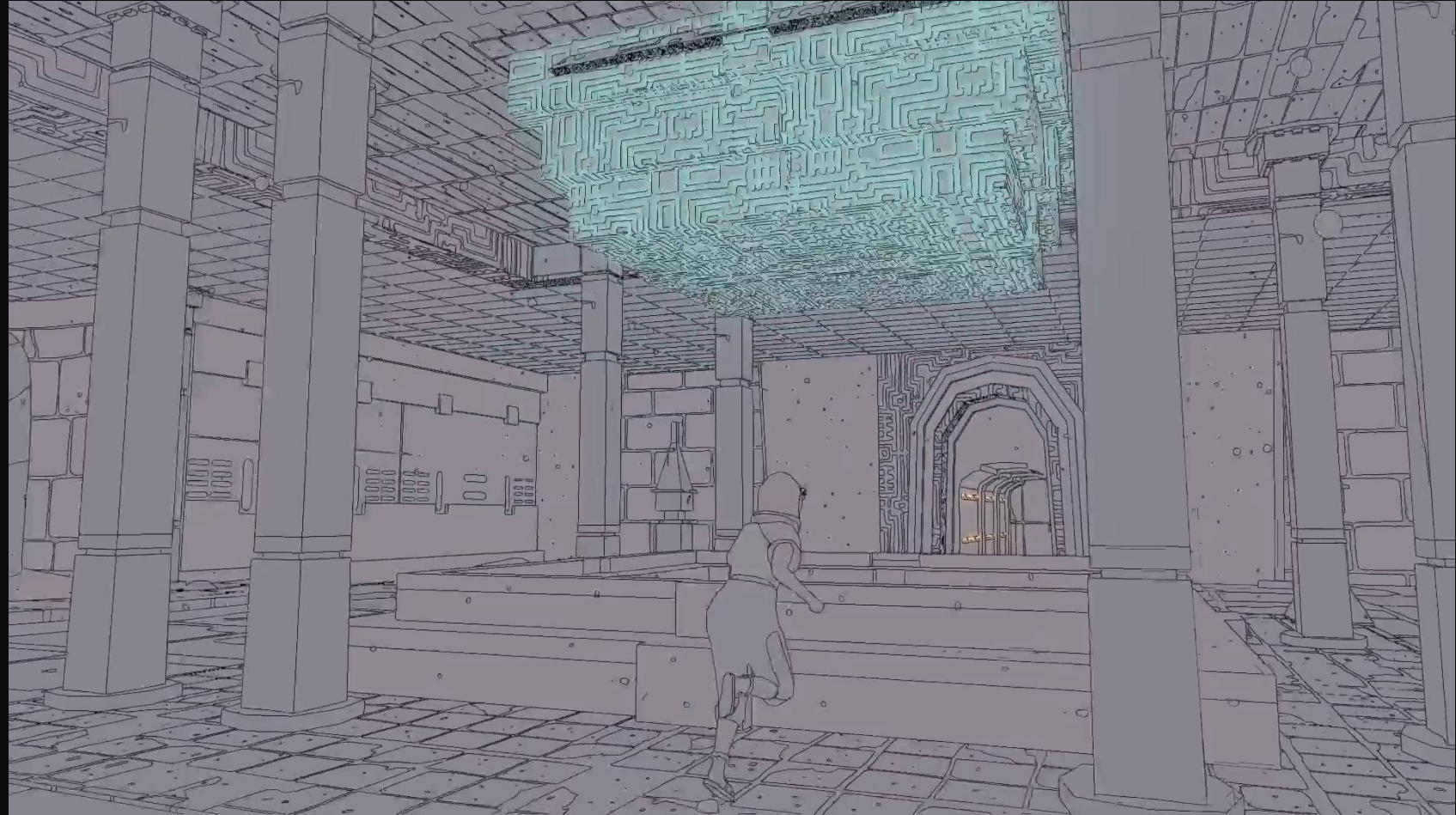




CAMERA PERSPECTIVE CONSIDERATIONS

Third Person Camera (Sable)

- Very popular for adventure games because you can see where you land when jumping.
- Shows character model so it is good for putting you into a specific character for more narrative type games.
- Harder to program a good camera than you think! (and a bad one will sink your game!)
 - Need to avoid walls etc.



SQUAD: VesuvianBow392 joined your squad

SQUAD: tomahawkredwolf left your squad



CrusaiderActual 1477

vagabond9595



Back Gunner

vagabond9595

0 2

CAMERA PERSPECTIVE: FIRST PERSON

First Person (Battlefield V)

- Very popular for shooters because you can aim easily (since there is no offset and the center of the screen can be used as a target reticle).
- YOU are the main character!
- Puts you directly in the action and people are used to seeing the world from this perspective.
- Needs some tricks to make it work (hint, you are just arms)



312

25:40

476

A B C D E F G

D

82 m
SWAMP

7 m
EIYak21

135 150 165

G C F E

SWAMP - NORTH

ATTACK
SWAMP

- USMC SEMPER FI6 15672
- ToK3Z
- EIYak21
- vagabond9595

54

1

1

1 + 100

60 m 6 30 2 397 m



EXAMPLES USED:

- Full Throttle – LucasArts (Tim Schafer of Doublefine)
- Cuphead – Studio MDHR
- Chrono Trigger - Square
- Hades – Supergiant Games
- Bad North – Plausible Concept (Published by Raw Fury)
- Sable – Shedworks (Published by Raw Fury)
- Battlefield V – Dice (EA)
- Doom – id software (Bethesda)